Dr. Jian Gu

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https://jianguda.github.io

EDUCATION

- **(Top 50) Monash University** — Melbourne, Australia
  PhD Degree (Supervised by Prof. Aldeida Aleti, Prof. Chunyang Chen, Prof. Hongyu Zhang) — *Oct 2022 – Oct 2026*
  - **Visiting Scholar**: NLP Lab @ THU (Oct 2024 – Jun 2025), SE&AI Lab @ TUM (Jun 2025 – Aug 2025)
- **(Top 100) KTH Royal Institute of Technology** — Stockholm, Sweden
  Master's Degree in Machine Learning (Thesis Advisor: Prof. Martin Monperrus) — *Aug 2018 – Aug 2020*
- **(Project 985) Shandong University (the Elite Program)** — Jinan, China
  Bachelor's Degree in Computer Science (Thesis Advisor: Prof. Jun Ma) — *Sep 2012 – Jun 2016*

SELECTED PAPERS

- **Lexical Reference Frames: Interpreting LM Residual Streams in Native Semantics**
  ***Jian Gu**, Hongyu Zhang, Chunyang Chen, Aldeida Aleti* — *Under Review*
- **Mind the Gaps: From Failure Attribution to Closed-Form Repair of Code LMs**
  ***Jian Gu**, Hongyu Zhang, Chunyang Chen, Aldeida Aleti* — *Under Review*
- **Rethinking Weight Tying: Pseudo-Inverse Tying for LM Stable Training and Updates**
  ***Jian Gu**, Aldeida Aleti, Chunyang Chen, Hongyu Zhang* — *Under Review*
- **Cross-Scale Knowledge Transfer in LMs through Latent Semantic Alignment** — ACL'26
  ***Jian Gu**, Aldeida Aleti, Chunyang Chen, Hongyu Zhang* — *(CORE A*)*
- **Semantic-based Optimization for Repairing LLMs: Case Study on Code Generation** — ICSE'26
  ***Jian Gu**, Aldeida Aleti, Chunyang Chen, Hongyu Zhang* — *(CORE A*)*
- **Semantic-Aware Layer-Freezing for Computation-Efficient Fine-Tuning of LMs** — ACL'25
  ***Jian Gu**, Aldeida Aleti, Chunyang Chen, Hongyu Zhang* — *(CORE A*)*
- **Vocabulary-Defined Semantics: Latent Space Clustering for Beyond-Context Learning**
  ***Jian Gu**, Aldeida Aleti, Chunyang Chen, Hongyu Zhang* — *Under Review*
- **Neuron Patching: Semantic-based Neuron-level LM Repair for Code Generation** — TOSEM 2026
  ***Jian Gu**, Aldeida Aleti, Chunyang Chen, Hongyu Zhang* — *(JCR Q1)*
- **Towards Top-Down Automated Development in Limited Scopes: A Neuro-Symbolic Framework from Expressibles to Executabless** — FSE'23 (IVR)
  ***Jian Gu**, Harald C. Gall* — *(CORE A*)*
- **Assemble Foundation Models for Automatic Code Summarization** — SANER'22
  ***Jian Gu**, Pasquale Salza, Harald C. Gall* — *(CORE A)*
- **Multimodal Representation for Neural Code Search** — ICSME'21
  ***Jian Gu**, Zimin Chen, Martin Monperrus* — *(CORE A)*
- **Source-Free Detection and Impact Analysis of Compiler Optimization in Mobile Apps** — ASE 2026
  *Han Hu, Xiaoheng Xie, Bo Sun, **Jian Gu**, Gang Fan, Li Li* — *(CORE A*)*
- **On the Effectiveness of Transfer Learning for Code Search** — TSE 2022
  *Pasquale Salza, Christoph Schwizer, **Jian Gu**, Harald C. Gall* — *(JCR Q1)*
- **Automated Classification of Overfitting Patches with Statical Code Features** — TSE 2021
  *He Ye, **Jian Gu**, Matias Martinez, Thomas Durieux, Martin Monperrus* — *(JCR Q1)*

WORK EXPERIENCE

- **Ant Group** — Hang Zhou, China
  Research Intern — *Feb 2026 – Aug 2026*
  - **LM Distillation**: proposed a novel distillation technique for LMs, enabling better support for edge-device tasks.
- **Huawei Research Center** — Hong Kong, China
  Research Intern — *Sep 2025 – Jan 2026*
  - **Byte-Level Encoder**: compiler provenance recovery for performance load reduction in HarmonyOS ecosystem.
  - **On-Device Small LMs**: Small LM pretraining and post-training for intelligent coding in HarmonyOS IDE.
- **Monash University** — Melbourne, Australia
  Teaching Assistant and Co-Supervisor — *Jul 2023 – Jun 2024*
  - **Teaching**: FIT5003 Software Security (1 semester); FIT4701/FIT4702 Projects and Research Theses (1 year).
- **University of Zurich** — Zurich, Switzerland
  Research Assistant, PhD Candidate — *Sep 2020 – Sep 2022*
  - **Research**: Conducted research on AI for SE, including code retrieval, comment generation, and code generation.
  - **Teaching**: Designed and advised "AI Pair Programming", a Master's team project on code intelligence (1 year).
- **KTH Royal Institute of Technology** — Stockholm, Sweden
  Research Engineer — *Jun 2019 – Aug 2019*
  - **Eclipse Repairnator**: Repairnator is a pioneering bot for automated program repair, hosted by the Eclipse Foundation. Integrated *Coming* and *SequenceR* into it and implemented its official GitHub app. Implemented *Prophet4J* (incl. *Sketch4Repair*) to generate code patches. Implemented *Context2Name* to prettify test cases.
- **Smart Software Studio** — Yancheng, China
  Software Engineer — *Jul 2017 – Jun 2018*
  - **YoYo**: Delivered a technical solution including websites and apps for online community, with the supports for local advertisements. It is for identity-based socials between alumni and neighbors, serving over 2k human users.
  - **WiiPM**: Developed a 2D rogue-like game on the mobile platform. WiiPM contains elements of turn-based battles as well as collection and role-plays adventure. The techniques of procedural content generation are adopted.
- **UCar Inc.** — Beijing, China
  Software Engineer — *Mar 2016 – Nov 2016*
  - **Anomaly Detection for Anti-Fraud**: Analyzed unusual operational logs, summarized anomaly patterns.
  - **Streaming Data Pipeline and Panel**: Maintained the streaming data pipeline for real-time transactions.
- **Hisense Group (R&D Center)** — Qingdao, China
  Algorithm Intern — *Sep 2015 – Feb 2016*
  - **Movie GenePool**: Improved collaborative filtering recommender systems. Built an unified knowledge base using plot/review tags to fuse data providers. Developed a workflow for movie metadata processing and demonstration.

SERVICES

- **Software Engineering** — ACM TOSEM, IEEE TSE (Reviewer)
- **Machine Learning** — AAAI (PC Member); ARR, ICLR, NIPS (Reviewer)
- **Data/Info Management** — CIKM (PC Member)
- **Multidisciplinary** — WWW (PC Member); IEEE TAI (Reviewer)

HONORS & GRANTS

- **Award for Outstanding Overseas Study Elite** — Chinese Government 2025 – 2026
- **Doctoral Scholarship, Top-Ups & Supplementary** — Monash University 2022 – 2026
- **$80k (2000 kSU) Research Grant** — AU National Computational Infrastructure 2024 – 2026
- **$30k HPC Research Grant** — DUG Technology Ltd 2023 – 2024
- **Ranked #1 (until Jun 2020)** — GitHub CodeSearchNet Challenge 2019 – 2020